

NAMED ACCOUNT / KOHLER INDIA CORPORATION / JHAGADIA

Cisco Enterprise Network

Micro-Level Technical Installation

Solution design, implementation, commissioning and operational handover.

01**CISCO PLATFORM**

Enterprise switching

02**POE ACCESS LAYER**

Power and uplinks

03**WI-FI 6 COVERAGE**

Wireless mobility

FIBRE

OM3 backbone

CAT6

User access

CCTV

Axis video

DISPLAY

Visual endpoints

CONVERGED INDUSTRIAL CONNECTIVITY AND SECURITY IMPLEMENTATION**3**

CATALYST 9200L
SWITCHES

7

CATALYST 9115AX APs

300 m

ARMoured OM3 FIBRE

50

STRUCTURED DATA
OUTLETS

D-TECH / DESIGN • DEPLOYMENT • CONFIGURATION • TESTING • DOCUMENTATION

Client-safe case study based on awarded technical scope PO 4200979707INJ3.

Connectivity, mobility and visibility

One engineered platform: D-Tech implemented a Cisco-led industrial network foundation integrating PoE switching, Wi-Fi 6, fibre backbone, structured cabling, Axis surveillance and operational displays at Kohler Jhagadia.

3

CISCO C9200L POE+ SWITCHES

7

CISCO C9115AX ACCESS POINTS

10

AXIS NETWORK CAMERAS

36 mo

CISCO SMART NET SUPPORT

Solution intent / reliable access, predictable coverage and a documented physical layer

The deployment established a manageable PoE access layer, industrial fibre interconnects and consistent user connectivity. A common network foundation supports surveillance and visual operations. Every rack, port, fibre core, outlet, access point and camera was mapped to a controlled work package.

01 Surveyed

Routes, racks, endpoint positions, fibre path and Wi-Fi coverage inputs.

02 Engineered

LLD, port plan, VLAN readiness, PoE capacity and installation method.

03 Implemented

Fibre, Cat6, racks, switches, APs, cameras, licenses and displays.

04 Validated

Physical layer, uplinks, PoE, RF, streaming, recording and handover.

DOCUMENTED PROJECT VALUE

INR 2.377 million gross | Approx. USD 0.029 million presentation equivalent at INR 83/USD.

Requirements & engineering basis

The named account required a coordinated active and passive infrastructure deployment for enterprise users, wireless mobility, IP surveillance and operational displays.

01 Managed PoE access

Three enterprise switches with power, uplinks and support coverage.

02 Wi-Fi 6 mobility

Seven access points with coverage, mounting and network readiness.

03 Industrial fibre path

Armoured OM3 backbone, LIUs, splicing and duplex patching.

04 Structured user access

50 Cat6 outlets with faceplates, back boxes and patch panels.

05 Rack discipline

Two 6U racks, cable managers, device mounting and clean dressing.

06 IP video integration

Eight bullet and two PTZ cameras with licenses and PoE support.

07 Display connectivity

Two 55-inch displays through controlled HDMI pathways.

08 Service continuity

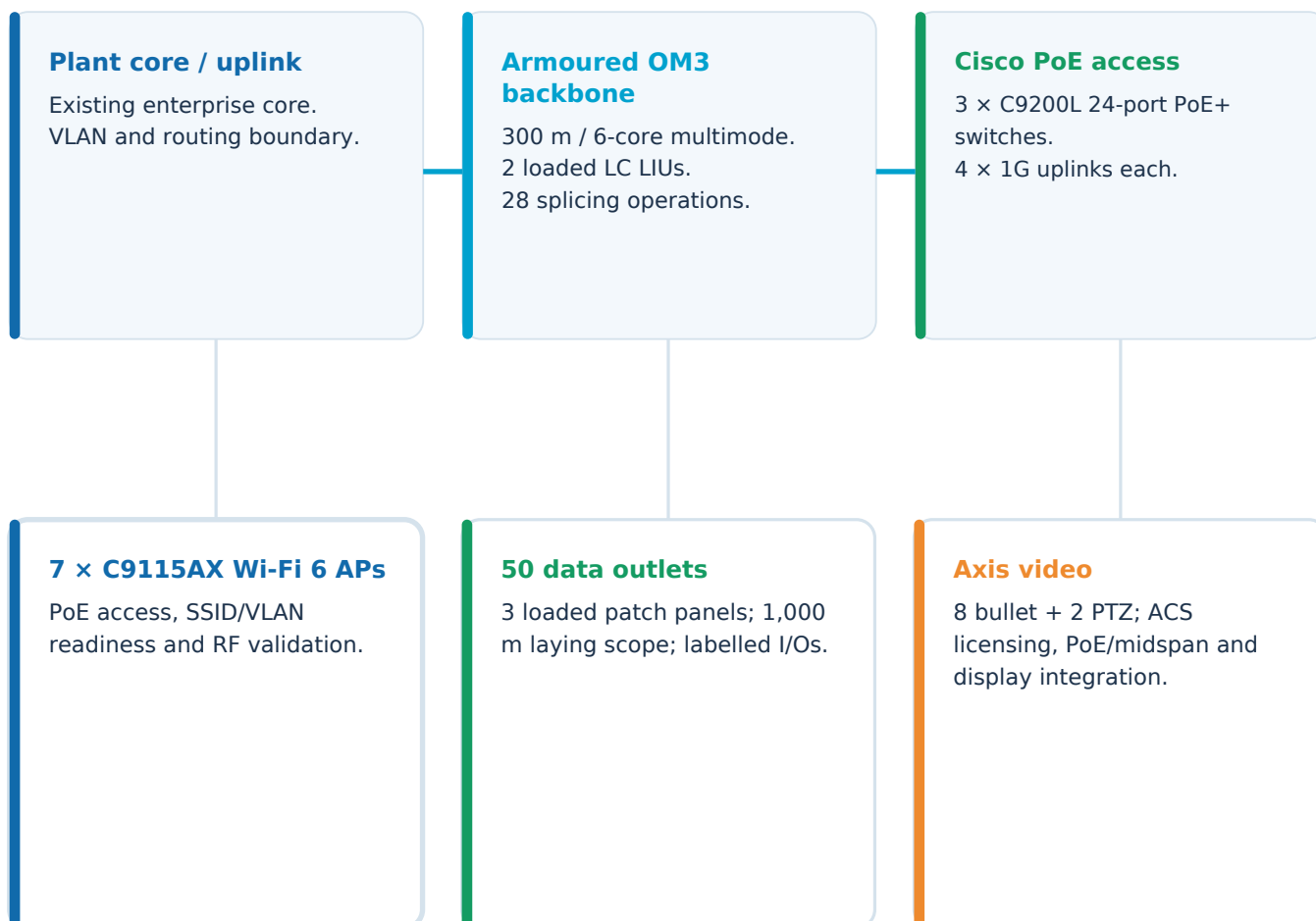
36-month Cisco support for switches and access points.

ENGINEERING PRINCIPLE

Every active device depends on a verified passive path, labelled port, known power source and documented logical assignment.

Solution architecture

Conceptual architecture showing the plant backbone, Cisco access switching, wireless coverage, user outlets, Axis surveillance and visual endpoints. Exact addressing and security rules remain client-confidential.



MANAGEMENT AND HANDOVER LAYER

Port map | VLAN readiness | RF checks | Fibre results | Device inventory | As-built records

Conceptual architecture based on the documented solution scope; not a construction drawing.

Full technical installation scope

Active, passive, wireless, video and display layers were implemented in one coordinated workstream.

SOLUTION LAYER	SCALE	TECHNICAL CAPABILITY
Cisco switching	3 × C9200L-24P-4G-E	24-port PoE+, four 1G uplinks, Network Advantage
Cisco wireless	7 × C9115AXI-D	Catalyst 9115AX Wi-Fi 6 access points
Cisco support	10 device contracts	36-month SNTC 8x5 next-business-day coverage
Optical uplinks	4 × GLC-LH-MMD	Cisco 1000BASE multimode fibre transceivers
Fibre backbone	300 m + 2 LIUs	6-core OM3 armoured fibre, LC patching and pigtails
Fibre services	28 splices	Cable laying, LIU installation, splicing and validation
Structured cabling	50 user points	Cat6 I/Os, faceplates, boxes and 3 loaded patch panels
Rack layer	2 × 6U racks	Wall-mount cabinets, managers, mounting and dressing
Axis surveillance	10 cameras	8 bullet, 2 PTZ, mounts, midspans and 10 ACS licenses
Visual endpoints	2 × 55-inch	Samsung displays with 15 m HDMI connectivity

Order reference Kohler India Corporation PO 4200979707INJ3 • 12 March 2024 • Jhagadia, Gujarat

Micro-level installation work pack

Every route, termination and endpoint was controlled through traceable work packages.

01 Pre-start survey

Freeze rack, AP, camera, display and outlet positions; confirm access, height and pathway constraints.

02 Material verification

Match model, quantity, serial, support entitlement, optics and accessories against approved scope.

03 Pathway preparation

Install 300 m HDPE path and 100 m casing with bends, supports, entry protection and route discipline.

04 Fibre installation

Pull 6-core OM3 armoured cable, maintain bend radius, protect ends and identify termination points.

05 LIU and splicing

Mount LIUs, prepare fibre, splice 28 cores, manage pigtails and route duplex LC patch cords.

06 Rack installation

Level, bond and secure two 6U racks; mount managers, patch panels and active equipment.

07 Cat6 termination

Lay approximately 1,000 m, fit 50 boxes and faceplates, punch I/Os and patch-panel ports.

08 Cisco device mounting

Install three switches and seven APs; verify power, uplinks, security and labels.

09 Endpoint integration

Install Axis cameras, PTZ mounts, midspans, licenses, displays and HDMI connections.

10 Testing and as-built

Record fibre, copper, PoE, RF and endpoint checks; update port map and handover schedule.

FIELD QUALITY RULE

No active endpoint is commissioned until mounting, cable path, termination, label, power and logical assignment are verified together.

Cisco configuration & commissioning

Structured configuration converts installed hardware into an operational access platform. IP addressing, VLAN IDs, SSIDs and credentials are intentionally omitted.

CONTROL POINT	MICRO-LEVEL TECHNICAL CHECK	OUTCOME
Asset baseline	Switch/AP model, serial, hostname and support entitlement	Inventory reconciled
Software readiness	Approved release, boot image, license state and time source	Stable baseline
Management plane	Management reachability, secure administration and logging	Controlled access
Layer-2 design	Access/trunk role, allowed VLANs, native VLAN and loop protection	Correct segmentation
PoE delivery	Port power state, class, consumption and total switch budget	Endpoint stability
Fibre uplinks	SFP seating, Tx/Rx status, speed/duplex and link errors	Clean backbone
AP onboarding	AP reachability, controller readiness and site/tag assignment	Managed wireless
RF profile	2.4/5 GHz policy, channel width, channel/power plan	Predictable coverage
SSID services	SSID-to-VLAN mapping, authentication and client addressing	Usable mobility
Operations	Backup, port map, configuration record and escalation path	Supportable handover

Cisco support foundation

Three switch and seven AP support contracts provide 36-month OEM service coverage for the active platform.

Testing, validation & documentation

Commissioning connects physical quality, active-network health, wireless experience and endpoint behaviour. Each check produces evidence for handover and future support.

01 Fibre continuity

Verify end-to-end cores, polarity, patching and acceptable optical condition.

02 Copper certification

Check pinout, wire map, length, termination quality and outlet identification.

03 Switch health

Validate boot, interfaces, uplinks, logs, errors, temperature and PoE state.

04 PoE load

Confirm connected device draw and available budget under expected load.

05 Wi-Fi service

Test association, authentication, DHCP, gateway access and roaming.

06 RF acceptance

Review coverage, signal quality, interference and channel utilisation.

07 Video stream

Validate image, PTZ control, time, recording, playback and licenses.

08 Display path

Confirm source input, HDMI continuity, resolution and stable operation.

09 Labelling audit

Reconcile rack, patch panel, switch port, outlet, AP and camera IDs.

10 Handover pack

Submit asset list, port map, fibre map, tests, support details and as-built.

ACCEPTANCE LOGIC

A service is accepted only when the passive route, active port, logical policy and endpoint experience pass as one chain.

Customer outcomes

A scalable plant network with clear ownership and serviceability. The solution improves endpoint power control, troubleshooting visibility, mobility, surveillance reach and technical records.

01 Enterprise access layer

Cisco switching provides a manageable PoE foundation for multiple IP services.

02 Wi-Fi 6 capability

Seven Catalyst APs extend wireless connectivity across key areas.

03 Resilient media

Armoured OM3 fibre creates a protected inter-rack backbone.

04 Port-level traceability

Mapped outlets and patch panels accelerate moves, changes and fault isolation.

05 Unified implementation

One partner coordinates active, passive, wireless, video and display layers.

06 OEM support coverage

36-month Cisco support protects the active-network investment.

07 Security visibility

Axis IP video uses the same disciplined network foundation.

08 Expansion readiness

PoE, fibre and cabling support future endpoints and new use cases.

SUCCESS MEASURE

A well-installed network is easy to operate, easy to audit and ready to expand without rebuilding the foundation.

Named account evidence & blueprint

This named-account engagement demonstrates coordinated design, installation, configuration, commissioning and documentation across a multi-OEM scope.

D-TECH DELIVERY CAPABILITY

Survey | LLD | Installation | Configuration | Testing | Handover

Cisco switching and Wi-Fi | Industrial fibre | Structured cabling | Rack integration | Axis surveillance | Visual endpoints | OEM support alignment | As-built documentation

EVIDENCE	REFERENCE	BASIS
Named account	Kohler India Corporation Pvt. Ltd.	Jhagadia, Gujarat
Award reference	PO 4200979707INJ3	12 March 2024
Core platform	Cisco Catalyst 9200L + 9115AX	Switching + Wi-Fi 6
Integrated scope	Fibre, Cat6, Axis CCTV and displays	Turnkey implementation
Gross value	INR 2.377 million	Approx. USD 0.029 million

Client-safe disclosure

This case study uses awarded scope and quantities. Confidential addressing, credentials, VLAN IDs, RF settings, serial numbers, contacts and unit prices are excluded. USD value is a rounded presentation equivalent.

Engineered in detail. Implemented with control. Ready for growth.